

Week 1: What is AI / ML?

AI Frameworks

Today's learning goals

By the end of this lecture you should be able to:

1. Tell apart **AI**, **ML**, and **data science** in plain words
2. Say what this course *will* and *won't* cover
3. Explain **supervised learning** with examples, labels, and predictions
4. Name our main tools: Python, Jupyter, and key libraries

Agenda (80 minutes)

1. Welcome and how the course works
2. AI vs ML vs data science
3. Scope: what we cover (and what we skip)
4. Supervised learning intuition
5. Tools overview: Python, Jupyter, libraries
6. Looking ahead to practice

Short quizzes are pauses to check understanding—not a test grade.

Who this course is for

- College students (about 16–18)
- Curious about AI, data, and building simple models
- Comfortable learning by running code and reading charts

Remember

We favour **intuition and practice** over heavy mathematics. You will see pictures and examples more often than formulas.

Weekly rhythm

Lecture — 80 min

- Ideas and demos
- Short quizzes
- Questions welcome

Practice — 80 min

- Guided notebooks
- Pair work
- Done checklist

Today after lecture: set up the environment and make your first predictions *by hand*.

Three overlapping ideas

Think of three circles that overlap—not three enemies.

AI

Systems that seem to “think” or act intelligently (broad umbrella).

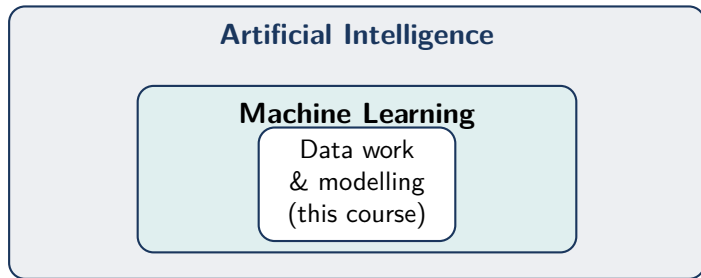
Machine learning

Programs that *learn patterns from data* instead of only hard-coded rules.

Data science

Asking questions of data: clean, explore, visualise, sometimes model.

A simple picture



Most of our semester lives inside the ML / data circle—practical modelling with tables.

Everyday examples (no math)

- **Recommendations:** “students who liked X also liked Y”
- **Spam filters:** learn from past spam vs not-spam emails
- **Photo tags:** the phone suggests who is in the picture
- **Weather apps:** models trained on historical measurements

Common pattern: **lots of examples** → **a system that generalises** to new cases.

Rules vs learning

Hand-written rules

- if score $< 50 \rightarrow$ fail
- Brittle when life is messy
- Easy to explain

Learning from data

- Show many examples
- Algorithm finds patterns
- Needs careful checking

This course teaches the second style—with tools that keep the workflow organised.

Quiz

Which statement is the best match for **machine learning**?

1. Designing web-page layout with HTML and CSS
2. Writing only fixed `if-else` rules with no data
3. Learning patterns from examples so we can make predictions on new cases
4. Replacing all of mathematics with ChatGPT

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

3. Learning patterns from examples to predict on new cases.

AI is the broad umbrella; data science often includes exploring and communicating with data; ML is specifically about learning from data.

What this course *will* cover

- Working with tables in **pandas**
- Charts with **matplotlib** (and a little seaborn)
- Feature types: numerical vs categorical
- Train / test thinking; simple metrics
- Models with **scikit-learn**: k-NN, linear & logistic regression, trees
- A mini-project and a first look at ethics / limits of AI

What we will *not* deep-dive

- Heavy calculus or “deriving gradients”
- Matrix theory and advanced linear algebra
- Building ChatGPT-scale models from scratch
- Transformer internals and research papers

Common mistake

Thinking you must be a maths olympiad winner to start. Curiosity + careful practice is enough for this course.

How success looks after a few weeks

You will be able to:

1. Open a CSV, explore it, and plot something meaningful
2. Explain features and a target in a supervised problem
3. Train a simple model and talk about whether it seems useful

Fancy architectures can wait for a later course.

Quiz 2 — Course scope (true/false)

Quiz

True or false?

In this course we will derive neural-network training maths (gradients) from scratch.

1. True
2. False

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

False. We stay practical: libraries, intuition, and clear experiments—not heavy derivations.

The core idea

Supervised learning

We have **examples** with known **answers** (labels). We want a model that predicts the answer for **new** examples.

- **Features / inputs:** what we know about each example
- **Label / target:** the answer we want to predict
- **Model:** the learned rule (often a function from features $\rightarrow$ prediction)

Two common flavours

Classification

Predict a **category**

spam / not spam

pass / fail

cat / dog / bird

Regression

Predict a **number**

house price

temperature

exam score

Same big idea; different kind of answer.

A tiny table (imagine a spreadsheet)

Size (m ²)	Rooms	City centre?	Price (k)
50	2	yes	180
80	3	no	210
65	2	yes	200
...	...	...	...

- Features: size, rooms, city centre
- Label: price
- Task type: **regression** (a number)

Training vs using the model

1. **Train:** show the model many examples with correct labels
2. **Predict:** give new features; get a predicted label
3. **Check:** compare predictions to reality on data the model did *not* train on

We will practise the “don't cheat with the test data” idea in a later week.

Not everything is supervised

Quick awareness (details later if at all):

- **Unsupervised:** find structure without labels (e.g. group similar customers)
- **Generative AI:** create text or images (powerful tools; different workflow)

This semester's backbone is **supervised learning on tables**.

Quiz 3 — Supervised learning

Quiz

You predict whether an email is spam or not spam from its words. What kind of supervised task is this?

1. Regression
2. Classification
3. Neither—there are no features
4. Unsupervised clustering only

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. Classification — the label is a category (spam / not spam).

If we predicted “how many spam emails arrive tomorrow” (a count), that would lean toward regression.

Quiz 3b — Features and labels

Quiz

In the housing table, **price** is best described as:

1. A feature used as input
2. The label / target we want to predict
3. The programming language
4. A type of chart

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. The label / target — what we try to predict from the other columns.

Our toolkit this semester

- **Python** — the language
- **Jupyter** (or Google Colab) — notebooks: text + code + results together
- **pandas** — tables (DataFrames)
- **matplotlib** — plots
- **scikit-learn** — classical ML models with a consistent API

Together, these are the “framework” feeling: shared patterns like `fit` / `predict`.

Why Jupyter?

- Run a small piece of code; see output immediately
- Mix explanations and results (great for learning and projects)
- Easy to share a `.ipynb` file or use Colab in the browser

Try this

In practice today: create a notebook, run a cell with `print("Hello, AI Frameworks!")`.

A first taste of Python in a notebook

```
1 print("Hello, AI Frameworks!")
2
3 # A tiny prediction "by hand" (no ML library yet)
4 size = 70                #  $m^2$ 
5 guess_price = 2.5 * size + 40    # made-up rule
6 print("Guessed price (k):", guess_price)
```

Later, libraries will learn better rules from real data—same idea, less guesswork.

Libraries = don't reinvent everything

- Professionals rarely write sorting, plotting, or k-NN from scratch in coursework
- Libraries give tested tools and a common language with teammates
- Your job: ask good questions, prepare data, interpret results

Common mistake

Thinking “using a library means I’m not really learning.” Using tools well *is* the skill—plus understanding what they do.

Quiz

Which tool is mainly for working with **tables** of data in Python?

1. matplotlib
2. pandas
3. HTML
4. Photoshop

Pause — answer (clicker / raise hand / neighbour) before the next slide.

Answer

2. pandas — DataFrames for tabular data.
matplotlib is for plots; we will use both often.

Key takeaways

1. AI is broad; ML learns from data; data science includes exploring data
2. This course is practical and light on formal maths
3. Supervised learning: features + labels $\rightarrow$ predictions on new cases
4. Classification predicts categories; regression predicts numbers
5. Python + Jupyter + pandas / matplotlib / scikit-learn are our tools

Next up

Practice (today)

Environment check · first notebook · predict house price by hand

Next lecture (Week 2)

Python enough for data work: lists, dicts, loops, functions

What is still fuzzy: AI vs ML, labels, or the tools?